

Problem description

Many animal shelters still manage adoptions using paper forms or simple spreadsheets. Information about animals, adopters, medical checks, and applications is often scattered, which makes it hard to track which animals are available, which adopters are approved, and how many adoptions succeed over time. A centralized database is needed to store and connect this information in a consistent way.

Users:

- *Adopters* – people who submit applications to adopt animals and provide their contact details
- *Shelter Staff* – employees or volunteers who register animals, update their status, and process adoption applications
- *Veterinarians* – users who record vet checks and medical treatments for animals
- *Shelter Managers* – users who oversee locations, staff, and reports (e.g., monthly adoptions, occupancy)

Entities:

- *Animal* – each pet in the shelter (name, species, breed, age, status, intake date, location)
- *Adopter* – a person interested in adopting (name, contact information, address)
- *Application* – a request from one adopter to adopt one specific animal, with status (pending, approved, rejected)
- *VetCheck* – a health check or treatment performed on one animal, recorded by one staff member or vet
- *Staff* – shelter workers or volunteers who manage animals, applications, and vet checks
- *ShelterLocation* – a physical shelter or foster location where animals and staff are assigned

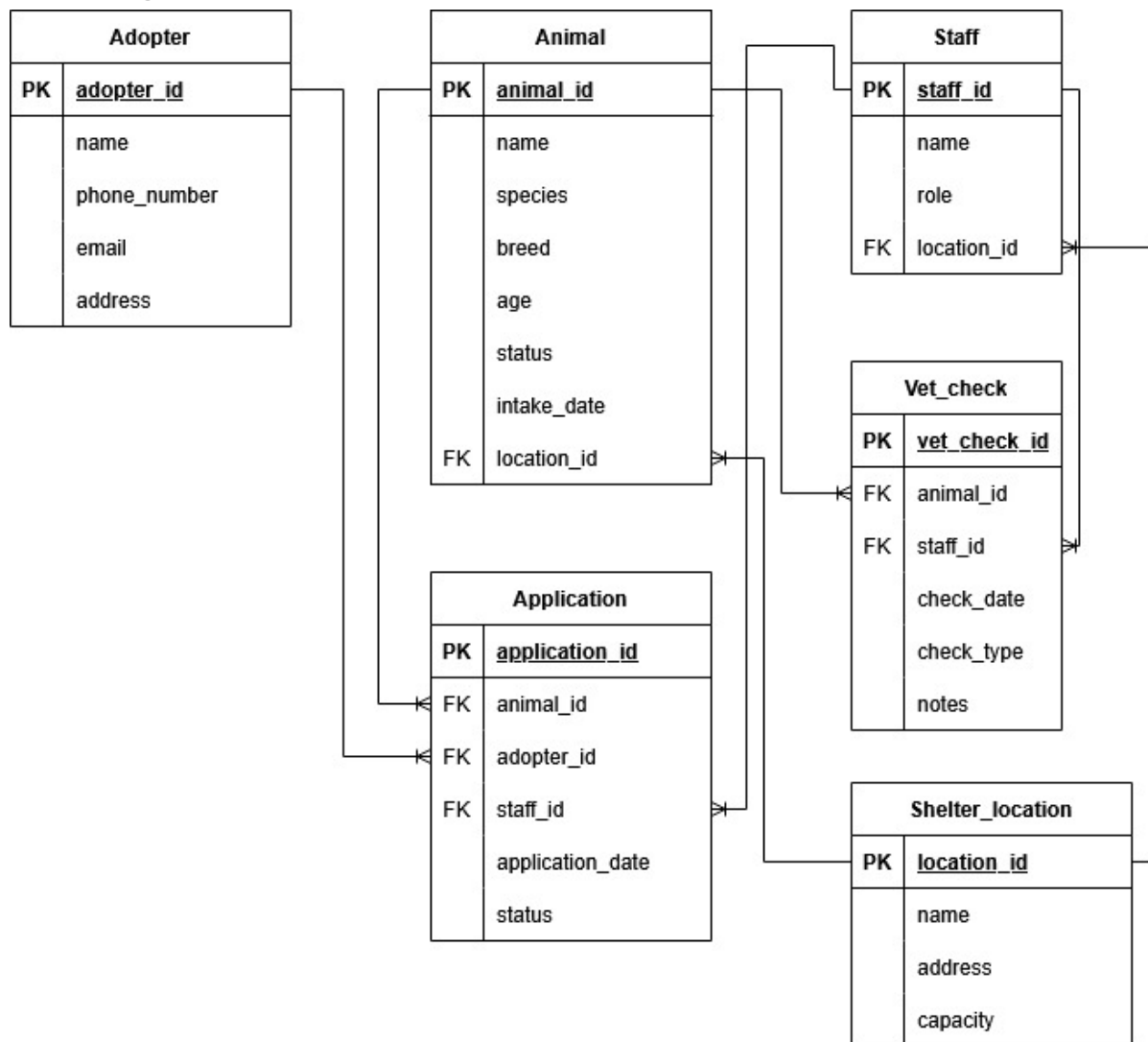
Relationships:

- One *Adopter* can submit many *Applications*; one *Animal* can receive many *Applications*
- One *Animal* can have many *VetChecks*
- One *Staff* member can record many *Applications* and *VetChecks*
- One *ShelterLocation* can host many *Animals* and employ many *Staff* members

Business Rules

1. Each animal must be registered at exactly one shelter location
2. Each adoption application must link exactly one adopter to exactly one animal
3. An animal may have many applications but can be adopted only once; after adoption its status becomes “Adopted”

ERD Diagram



Screenshots & Comments

1. Populated Table (Staff)

staff_id	name ↑	role	location_id
abc Filter...	abc Filter...	abc Filter...	abc Filter...
1	Anna Tšigrinski	Manager	1
2	Kyle Maclachlan	Veterinarian	1
3	Tobey Maguire	Veterinarian	1
4	Marcia Cross	Staff	1
5	Robert Pattinson	Manager	2
6	Ian Somerhalder	Veterinarian	2
7	Taylor Lotner	Staff	2
8	Zendaya	Manager	3
9	Tom Holland	Veterinarian	3
10	Jacob Batalon	Staff	3

2. This query uses INNER JOINS to list all approved adoption applications. It shows which animal was adopted, who adopted it, and which staff member processed it.

application_id	application_date	animal_name	adopter_name	staff_name
abc Filter...	abc Filter...	abc Filter...	abc Filter...	abc Filter...
1	2024-11-15	Bella	Jake Gyllenhaal	Anna Tšigrinski
2	2023-02-02	Kira	Tom Felton	Marcia Cross
4	2018-03-03	Hosico	Leonid Catopyesov	Zendaya

3. This query uses LEFT JOIN and GROUP BY to count animals in each shelter, including how many are available for adoption.

location_id	location_name	total_animals	available_animals
abc Filter...	abc Filter...	abc Filter...	abc Filter...
2	Hoiupaiga loomad MTÜ	2	2
3	Tartu Koduta loomade varjupaik	1	0
1	Varjupaikade MTÜ	2	0

4. This query ranks adopters by the number of approved adoptions using the RANK() window function. It helps the shelter identify frequent adopters.

adopter_id	name	approved_count	adoption_rank
abc Filter...	abc Filter...	abc Filter...	abc Filter...
1	Jake Gyllenhaal	1	1
2	Tom Felton	1	1
3	Leonid Catopyesov	1	1